S.D.C. HACKATHON 2026 · DOMAIN 2 · ORBYTE SMART SME CHALLENGE

Orbit

Stop chasing receipts on WhatsApp.

Payment reconciliation for the Orbyte hub: receipt in, invoice matched, owner approves, ledger updated.



Team: Atlas

github: https://github.com/Reqqa/Orbit
video: https://youtu.be/oLzOzDHE0Qs
prototype: https://orbit-casimirchiong.replit.app

Team Information

Who is behind Orbit.

Members (2–5)

Member	Name and role
Member 1 (Leader)	Tharmunieshwar / Lead AI Engineer
Member 2	Casimir Chiong / Implementation
Member 3	Shau Loon / Backend Engineer
Member 4	Gwee Jee See / Video and Slides

Delete any unused member. One person can hold two roles.

1. The Problem & Target User

Malaysian SMEs get paid through WhatsApp screenshots, and the owner reconciles each one by hand.

2–4 min
per receipt: open the chat, read it, check the invoice, key it in, reply

4–8 h
per week on reconciliation and chasing late payers

1 wrong key-in
can go unnoticed for a month

Assumed manual baseline, to be validated in a pilot. Demo data is synthetic.

WHO WE BUILD FOR

Owners
Approves what matters, sees everything.

Finance staff
Approves routine matches up to a limit.

Sales staff
Sees own customers only. Cannot approve.

Customers
Sends the receipt on WhatsApp. No dashboard.

2. Solution & Innovation

AI only reads the receipt. Rules decide. A person approves.

	Today: WhatsApp plus spreadsheet	With Orbit
Read the receipt	Owner squints at a screenshot	AI reads the image, nothing more
Match to an invoice	Eyeball plus spreadsheet	100-point scorecard, with reasons
Record the payment	Owner keys it in	One-tap approval by the right role
Chase late payers	Ad hoc, easy to forget	Reminder ladder, batch-approved
Keep a record	Spreadsheet edits	Hash-chained, tamper-evident log

A workflow product, not a chatbot. It adds receivables, not Orbyte's existing HR and admin functions.

3. How it Works

From a WhatsApp screenshot to a recorded payment, with a person approving.

1

Trigger

Customer sends a receipt on WhatsApp.

2

Connected data

Open invoices, customers and roles come through the Orbyte connector.

3

Decision

One AI call reads the image. A rules scorecard matches and classifies it.

4

Approval

The right role approves with one tap. Limits are enforced on the server.

5

Outcome

Invoice marked paid, customer notified, audit log updated.

Built with Python 3.12 · FastAPI · SQLite · Next.js · Tailwind · Claude vision, swappable behind one interface.

The demo runs on a built-in WhatsApp simulator that posts the real Cloud API payload shape. It is labelled as simulated on screen.

4. People in Control & Orbyte Fit

Roles, approval and an audit trail, on top of a shared business hub.

People stay in control

- Finance approves clean matches up to RM 2,000; the Owner decides the rest
- Sales sees own customers and is refused (403, logged)
- Nothing is recorded until a person approves
- Audit log is hash-chained, with verify and export
- PDPA: keyed phone hash, retention limits, synthetic data only

Orbyte fit and reuse

- Connector is on the live path: every receipt is scored against invoices it returns
- Every approval writes a payment and an activity event back
- Seven assumed endpoints; no public Orbyte API exists, so a mock stands in
- Same code serves 3 tenants: trader, bakery supplier, tuition centre
- Reusable: POST /v1/receipts/analyze

5. Impact

Fewer manual steps, every payment human-approved. Supports SDG 8.2, 8.3 and SDG 9.

Measured on 50 synthetic receipts

47/50
classification accuracy (94%)

0
confident wrong matches

44%
auto-matched at score 90+; the rest go to a person

Extraction is 100% only by construction: the mock reads the labels. It is not a model measurement.

Targets, not yet measured

~3 min → ~20 s
per receipt

~6 h → ~1 h
per week on reconciliation and chasing

100%
of overdue invoices touched on the ladder

Against an assumed manual baseline. A timed pilot is next.

6. Feasibility

The core journey runs end to end on synthetic data, with its hardest risk named.

Built and working

- Webhook to extraction, match, approval, ledger, audit
- Chasing ladder with batch approval
- 3 tenants, 75 customers, 80 invoices, 50 receipts
- 40 automated tests, 25 documented API paths

Hardest risk

- WhatsApp Business API access (Meta verification)
- Cloud API sender is written but never run against Meta
- Fallback: simulator with the same payload, so only transport is stubbed
- Live reminders need approved templates

Next, stated plainly

- Pilot with 3–5 real SME owners
- Real Meta test number
- HTTP Orbyte connector if access is given
- Daily scheduler for chasing
- Not claimed: forged-receipt detection. Orbit never moves money.

7. Implementation & Growth

Prove it with a few SMEs, then widen what Orbit reconciles and where.

PHASE 1

Malaysian SMEs

Reconciliation and chasing, piloted with 3–5 businesses that invoice on credit.

PHASE 2

More signals

Bank statement matching, e-invoice readiness checks (preparatory), cash-flow forecast.

PHASE 3

ASEAN

Local receipt formats and languages: PayNow, QRIS, PromptPay.

Business model

Freemium (first 30 receipts a month free), then per-tenant tiers by receipt volume, plus partner revenue share through Orbyte with QR payment and e-invoicing providers.

8. Why Orbit Should Win

One recurring burden removed for Malaysian SMEs, with people in control.

Live
core journey, end to end

0
confident wrong matches on 50 receipts

7
Orbyte endpoints behind one connector

Thank you